Reflected Barnes saddles and pole oscillations in the recoil graph

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Abstract

We analyze the ordinary-Gamma internal integral obtained after the fixed-mass pure-recoil limit. Reflection exposes a quadratic saddle equation with discriminant equal to minus nine times the squared volume of the dual flat tetrahedron, retaining both transverse energies. The local Gaussian determinant reproduces the expected geometric factor, conditionally on its contribution to the full contour. An exact oscillating pole factor need not tend pointwise to one in the heavy-observer window. A uniform Fourier estimate explains why this alone does not invalidate an integrated saddle approximation. We distinguish exact algebra, numerical action comparisons, and the global contour and uniform estimates that remain open.

1 The exact reflected integrand

Set the de Sitter radius to one. First take $\kappa \to \infty$ at fixed positive real masses, as in the preceding internal-integral note. The object is the internal chiral amplitude of the fixed-topology CLP graph [2], not yet a normalized physical OTOC. Let

$$\mathfrak{B} = \frac{1}{2\pi i}\int_{\mathbb{R}-ic} \mathcal{B}(z)\,\mathrm{d}z, \qquad \mathcal{B}(z) = e^{-i\pi z}\frac{\prod_a \Gamma(z+ia)\prod_r \Gamma(\frac{1}{2}-z-ir)}{\Gamma(z+\frac{1}{2})},$$

where there are four a's and three r's, $\sum a = \sum r$, and $\max a < c < \min r$. The equal-probe parametrization retains both transverse energies:

$$m_{1,3} = M + \omega/2 \pm u, \qquad m_{2,4} = M - \omega/2 \pm v, \tag{1}$$

$$a = M + \nu/2 + (\pm u \pm v)/2, \qquad r = \{M+\nu+\omega/2,\ 2M,\ M+\nu-\omega/2\}. \tag{2}$$

The edge labels are $(\mu_1,\dots,\mu_6) = (m_1,\nu,m_2,m_3,\nu,m_4)$.

Reflection of the three right Gamma factors gives the exact identity

$$\mathcal{B}(z) = 8\pi^3 e^{2\pi i z - \pi\sum r}\frac{\prod_a \Gamma(z+ia)}{\Gamma(z+\frac{1}{2})\prod_r \Gamma(z+\frac{1}{2}+ir)}E(z), \qquad E(z) = \prod_r (1+e^{2\pi i(z+ir)})^{-1}. \tag{3}$$

For complex r, the horizontal line requires $c < \min \Re r$; the left poles must still stay above it. The factor E must not be discarded merely by invoking Stirling. In fact a principal-log expansion of the unreflected expression can give derivative $-2\pi i$, rather than zero, at the desired exponentiated root.

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2 Quadratic saddle and the local determinant

Write $f(t) = t(\log t - 1)$. On the branch continued through the right saddle, the reflected Stirling exponent is

$$\Phi(z) = 2\pi i z - \pi \sum r + \sum_a f(z + ia) - \sum_r f(z + ir) - f(z). \tag{4}$$

Exponentiating $\Phi' = 0$ gives $\prod_a (z + ia) = z \prod_r (z + ir)$. In the coordinate $s = iz$,

$$P(s) = \prod_a (s-a) - s \prod_r (s-r) = As^2 + Bs + C, \quad A = e_2(a) - e_2(r),\ B = -e_3(a) + e_3(r),\ C = \prod a. \tag{5}$$

Here e_k denotes an elementary symmetric polynomial. Balance cancels the cubic term; the quartic terms already cancel. In particular, $A = M^2 + \nu^2/2 - (u^2 + v^2)/2 + \omega^2/4$.

Let CM be the Cayley–Menger matrix of the auxiliary flat tetrahedron. Its edges are the complementary entries of the face-labeled mass matrix in CSTY (2.54), not that matrix interpreted as vertex distances [1]. Direct polynomial expansion gives

$$\Delta = B^2 - 4AC = -\frac{\det \mathrm{CM}}{32} = -9V_{\mathrm{dual}}^2. \tag{6}$$

This identity retains u, v. On their zero slice,

$$\Delta = -\frac{(\nu^4 - 4M^2\omega^2)(4M^2 - 2\nu^2 + \omega^2)}{16}. \tag{7}$$

For a nondegenerate real Euclidean configuration choose

$$s_+ = \frac{-B + i\sqrt{-\Delta}}{2A}, \qquad z_+ = -is_+, \qquad \sqrt{-\Delta} > 0,$$

and continue this branch. Calling a principal $\sqrt{\Delta}$ directly is numerically unsafe on its negative-real cut: a rounding-size imaginary part can select the other root.

Put $N = \prod_a (z_+ + ia)$. Differentiating the logarithmic saddle equation, or differentiating $P(iz)$, gives exactly

$$\Phi''(z_+) = -\frac{\sqrt{-\Delta}}{N}. \tag{8}$$

The non-exponentiated logarithmic equation must still be checked. An algebraic root does not determine a thimble coefficient.

The half-integer shifts in (3) give the local factor $g = 8\pi^3 \prod_a (z + ia)^{-1/2}$. If the right saddle supplies the leading full-contour contribution and the pole factor may be replaced by its zero Fourier mode in that contribution, then for real masses

$$|\mathfrak{B}_{\mathrm{loc}}|^2 \sim \frac{|g(z_+)|^2}{2\pi |\Phi''(z_+)|} e^{2\Re\Phi(z_+)} = \frac{32\pi^5}{\sqrt{-\Delta}} e^{2\Re\Phi(z_+)}. \tag{9}$$

This is a conditional local Gaussian expression, not a theorem for the complete Barnes integral.

3 Comparison with the worldline action

CSTY's classical action is $I = \sum_{i<j} m_{ij} d_{ij}$, with the angles specified by the dual flat tetrahedron in Appendix B.1 [1]. Five finite-point evaluations, including nonzero u, v and imaginary ω, give

$$-2\Re\Phi(z_+) = I.$$

Three transverse Hessian comparisons also agree with

$$\partial_{u,v}^2 I\big|_{u=v=0} = h(\omega)\mathbf{1}_2, \qquad h(\omega) = \frac{4\sqrt{(\nu^4 - 4M^2\omega^2)(4M^2 - 2\nu^2 + \omega^2)}}{(4M^2 - \nu^2)(\nu^2 - \omega^2)}.$$

These are numerical consistency checks, not a supplied general proof of the angle identity. The imaginary-axis tests exploit the symmetries of this parametrization; they do not prescribe taking a modulus anew at arbitrary complex masses.

Inside the strict mass triangles, the preceding combined-vertex note gives the leading four-vertex factor

$$\prod_v T_v \sim \frac{\nu^2 m_1 m_2 m_3 m_4}{64\pi^6}.$$

Multiplying it with (9) gives the local graph coefficient

$$\frac{\nu^2 m_1 m_2 m_3 m_4}{2\pi\sqrt{-\Delta}}. \tag{10}$$

Thus the energy-dependent ratio is the previously found mass polynomial times $\sqrt{\Delta(0)/\Delta(\omega)}$. This derives the geometric determinant within the same internal representation. No vertex factor has been multiplied onto an already complete $6j$ approximation. Physical insertion normalization and external energy cycles remain separate questions.

4 Pole distance and a uniform oscillation estimate

On $u = v = 0$, write $c = -\Im z_+$. For real ω in the Euclidean region, and also on the symmetric imaginary-axis continuation, direct algebra gives

$$\delta = M + \nu - c = \frac{\nu(\nu^2 + \omega^2)}{4M^2 + 2\nu^2 + \omega^2}. \tag{11}$$

For real ω, the nearest right-pole row has height $\delta - |\omega|/2$ relative to this line, and it need not be positive. One must not move the original contour to that line across poles. On $\omega = iA_0 y$, $A_0 = \nu^2/(2M)$, bounded y, and $\nu/M \to 0$, both near rows instead have positive height $\delta \sim \nu^3/(4M^2)$; the third is farther away. There

$$|E - 1| \le (1 - e^{-2\pi\delta})^{-3} - 1. \tag{12}$$

Consequently $\nu^{4/3} \ll M \ll \nu^{3/2}$ is a sufficient subwindow for pointwise deletion of E. It is not established as a necessary window for an integrated asymptotic. At $M \asymp \nu^{3/2}$ the pole factor remains oscillatory, and for faster M it can be sharply peaked.

Lemma 1 (Uniform integrated pole-factor bound)**.** *For $|q_\rho| < 1$, let $E_q(x) = \prod_{\rho=1}^{3}(1+q_\rho e^{2\pi i x})^{-1}$. For every complex $F \in W^{4,1}(\mathbb{R})$,*

$$\left|\int_{\mathbb{R}} F(x)(E_q(x) - 1)\,\mathrm{d}x\right| \le C_4 \|F^{(4)}\|_1, \qquad C_4 = \frac{\zeta(2) + 3\zeta(3) + 2\zeta(4)}{2(2\pi)^4}. \tag{13}$$

The constant is independent of the distance of the q_ρ's from the unit circle.

Proof. For each fixed triple, expand the three geometric series:

$$E_q = 1 + \sum_{n\ge 1} c_n e^{2\pi i n x}, \qquad |c_n| \le \binom{n+2}{2}.$$

Four integrations by parts give

$$\left|\int F(x) e^{2\pi i n x}\,\mathrm{d}x\right| \le (2\pi n)^{-4} \|F^{(4)}\|_1.$$

The sum of $\binom{n+2}{2}/n^4$ is $(\zeta(2) + 3\zeta(3) + 2\zeta(4))/2$. For fixed q_ρ the initial exchange is justified by absolute convergence; the resulting summable majorant is uniform. □

On $z = x - ic$, take $q_\rho = e^{-2\pi(r_\rho - c)}$. The endpoint $|q_\rho| = 1$ is excluded; the lemma does not define an integral through poles. Normalize the actual localized amplitude by its nonzero saddle prefactor $g(z_+)$, and call the result F (excluding E). If it obeys

$$\|F^{(4)}\|_1 \le Ce^{\Re\Phi(z_+)}A_0^{-3/2}, \qquad \left|\int F\right| \ge c_0 e^{\Re\Phi(z_+)}\sqrt{A_0},$$

then (13) implies a relative $O(A_0^{-2})$ pole correction, uniformly as $\delta \downarrow 0$. These two estimates remain to be proved for the actual localized Gamma quotient. They are not consequences of simply naming a Gaussian saddle. The global quotient has not been given such a derivative bound either. This distinction prevents a pointwise Stirling failure from being mistaken for a failure of the integral.

5 Validation and outstanding work

The new script `code/recoil_saddle.py` records 57 passing checks: four symbolic identities; five groups of saddle, determinant, action, reflection and precision comparisons; nine transverse derivatives; eighteen scaling/precision checks; and one diagnosis of the failed unreflected principal-log shortcut. Calculations use 80 digits with specified 110-digit repeats. Nine growing-mass samples separately record the exact pole factor and the remaining Stirling error.

The initial implementation used a square root on its cut, selecting the wrong root in some samples. Its failed output is retained. After transporting $\sqrt{-\Delta}$, a largest-mass comparison also required more than 60 digits to meet the stated tolerance. Finite-point checks neither certify a global asymptotic nor establish priority. The algebraic quadratic and geometric discriminant have antecedents in the full $6j$ analysis [2].

The next tasks are a general angle-identity proof, localized uniform derivative and non-cancellation bounds, and a deformation of the full cycle that controls the other saddle and tails. The u, v exterior, two chiral continuations, physical insertion factors, three external energy cycles, and disconnected baseline remain unresolved. The fixed-mass κ limit has not been made uniform in growing masses. Finite-G dynamics and a complete de Sitter dual remain open.

References

[1] Y. Chen, D. Stanford, H. Tang and Z. Yang, *Negative shocks versus static patch holography*, arXiv:2607.14042v1, (2.53)–(2.54), Appendix B.1.

[2] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1, (3.46), Appendix B.